

Chapter 5, Problem 4P

Problem

The output voltage of an op amp is -4 V when the noninverting input is 1 mV . If the open-loop gain of the op amp is 2×10^6 , what is the inverting input?

Step-by-step solution

Step 1 of 3

Given that,

Output voltage $(V_o) = -4\text{ V}$

Non-Inverting terminal voltage $(V_2) = 1\text{ mV}$ and

Gain $(A) = 2 \times 10^6$.

Step 2 of 3

Therefore,

Output voltage $V_o = AV_{\alpha}$

$$V_o = A(V_2 - V_1)$$

$$V_2 - V_1 = \frac{V_o}{A}$$

$$V_2 - V_1 = \frac{-4}{2 \times 10^6}$$

$$V_2 - V_1 = -2\mu\text{V}$$

Step 3 of 3

$$1 \times 10^{-3} - V_1 = -2 \times 10^{-6}$$

$$-V_1 = -2 \times 10^{-6} - 1 \times 10^{-3}$$

$$-V_1 = -1.002\text{ mV}$$

$$\boxed{V_1 = 1.002\text{ mV}}$$